SET – 2

Git:

Step 1: Create HTML code for the given scenario

Step 2: git init

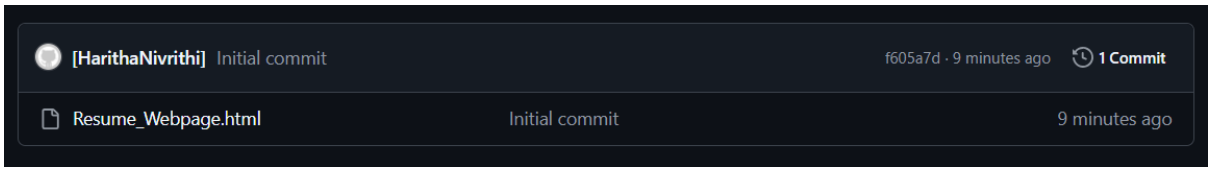
Step 3: git add .

Step 4: git commit -m "Initial commit"

Step 5: git branch -M main

Step 6: git remote add origin https://github.com/HarithaNivrithi/Agile-Set-B.git

Step 7: git push -u origin main



Step 8: git clone https://github.com/HarithaNivrithi/Agile-Set-B.git

Step 9: cd Agile-Set-B (Use the name of your git repository)

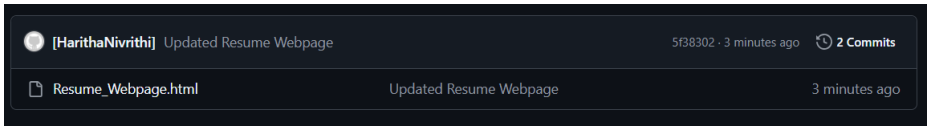
Step 10: code . (It'll open your folder and code in a new window in VS, Continue to work in the new window's terminal in VS)

Step 11: Make the changes asked according to the question

Step 12: git add .

Step 13: git commit -m "Updated Resume Webpage"

Step 14: git push



Docker:

Step 15: type null > Dockerfile

Step 16: Paste the code given for Dockerfile (From any AI)

```
Dockerfile > ...
# Use nginx web server
FROM nginx:alpine

# Copy all files (including index.html) to nginx folder
COPY . /usr/share/nginx/html

# Expose port 80
EXPOSE 80
```

Step 17: docker build -t harithanivrithi/resume-app . (Must include your docker hub username. Example: harithanivrithi/example)

Step 18: Check Docker Hub -> Images and Builds for the output. Also run (docker images) command in the vs terminal for terminal confirmation.

Step 19: docker push harithanivrithi/resume-app (Pushes image into Docker Hub Cloud)

Step 20: docker run -d -p 8081:8080 harithanivrithi/resume-app

Step 21: https://localhost:8081 (To open HTML webpage)

Jenkins:

Step 22: Download all necessary plugins – docker, Kubernetes.

Pipeline Script:

```
pipeline {

    agent any

    stages {

        stage('Clone') {
            steps {
                git branch: 'main', url: 'https://github.com/HarithaNivrithi/Agile-Set-B.git'
            }
        }

        stage('Build Docker Image') {
            steps {
                bat 'docker build -t resume-app .'
            }
        }

        stage('Run Container') {
            steps {
                bat 'docker run -d -p 8082:80 resume-app'
            }
        }
```

```
    }
}
```

Step 23: Build and ss output

Kubernetes:
Step 24: type null > deployment.yaml

Step 25: Paste the code given for Deployment.yaml (From any AI)

```yaml
deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: resume-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: resume
  template:
    metadata:
      labels:
        app: resume
    spec:
      containers:
      - name: resume-container
        image: harithanivrithi/resume-app
        ports:
        - containerPort: 80
```

Step 26: type null > service.yaml

Step 27: Paste the code given for service.yaml (From any AI)

```yaml
service.yaml
apiVersion: v1
kind: Service
metadata:
  name: resume-service
spec:
  type: NodePort
  selector:
    app: resume
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
```

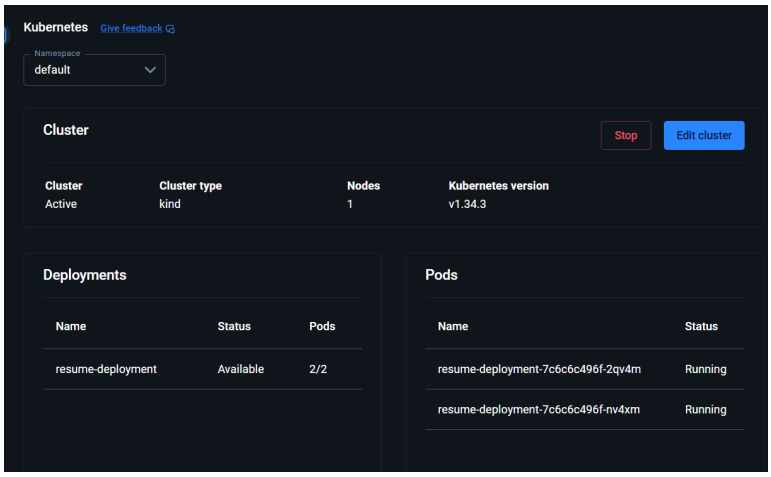
Step 28: kubectl apply -f deployment.yaml

Step 29: kubectl apply -f service.yaml

Step 30: kubectl get pods (To get Kubernetes output)

```
NAME                                  READY   STATUS    RESTARTS   AGE
resume-deployment-7c6c6c496f-2qv4m    1/1     Running   0          9s
resume-deployment-7c6c6c496f-nv4xm    1/1     Running   0          3s
```

Step 31: Kubernetes output from Docker Hub:



If ErrImage or status 0/1 occurs:

```
PS C:\Users\harit\OneDrive\Documents\Agile\Agile-Set-B> docker images
                                                                  i Info →   U   In Use
IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/desktop-cloud-provider-kind:v0.5.0      4ad59ce20658       595MB         162MB      U
docker/desktop-containerd-registry-mirror:v0.0.3
                                               179a9ea3ae47      50.3MB        16.1MB      U
envoyproxy/envoy:v1.36.4                       08acc53db6f4       248MB        62.1MB
harithanivrithi/resume-app:latest              ad1ce6481720      92.9MB          26MB      U
kindest/node:v1.34.3                           08497ee19eac      1.35GB         395MB      U
resume-app:latest                              b1873a2bee02      92.9MB          26MB      U
PS C:\Users\harit\OneDrive\Documents\Agile\Agile-Set-B> docker tag resume-app harithanivrithi/resu
me-app
PS C:\Users\harit\OneDrive\Documents\Agile\Agile-Set-B> docker login
Authenticating with existing credentials... [Username: harithanivrithi]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
PS C:\Users\harit\OneDrive\Documents\Agile\Agile-Set-B> docker push harithanivrithi/resume-app
Using default tag: latest
The push refers to repository [docker.io/harithanivrithi/resume-app]
f03becc8ac15: Layer already exists
0c84d000a0cf: Pushed
ff9f59a6a62e: Layer already exists
34dfdd2ef1f9: Layer already exists
15e759724ff6: Layer already exists
c8a2fa3a88d2: Layer already exists
1165b869c51a: Layer already exists
a71873b303e8: Layer already exists
4da5878fe1cc: Pushed
589002ba0eae: Layer already exists
latest: digest: sha256:b1873a2bee022bd561d516a90ccf79b276adc2f943375ccd73a9a0ad4ffe5103 size: 856
PS C:\Users\harit\OneDrive\Documents\Agile\Agile-Set-B> kubectl apply -f deployment.yaml
deployment.apps/resume-deployment configured
PS C:\Users\harit\OneDrive\Documents\Agile\Agile-Set-B> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
resume-deployment-7c6c6c496f-2qv4m  1/1     Running   0          9s
resume-deployment-7c6c6c496f-nv4xm  1/1     Running   0          3s
```